

GUNNISON-CRESTED BUTTE, CO, USA

WMO: 724677

Lat: 38.533N

Lon: 106.933W

Elev: 7666

StdP: 11.06

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93007

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-10.9	-26.2	1.7	-12.1	-21.6	2.2	-9.4	19.7	29.0	17.5	29.4	1.2	70	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.9	82.6	53.5	81.3	53.1	79.1	52.6	57.6	72.9	56.5	71.8	55.4	70.7	9.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.6	81.5	61.1	51.9	76.4	60.8	49.6	70.2	61.0	28.3	73.8	27.5	69.9	26.7	70.4	62.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.6	16.3	DB	-23.3	86.8	5.6	2.3	-27.4	88.5	-30.7	89.8	-33.9	91.1	-38.0	92.7	(4)
(5)			WB	-22.2	59.4	4.9	1.7	-25.7	60.6	-28.6	61.5	-31.3	62.4	-34.9	63.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	11.9	16.8	30.0	39.6	48.5	57.6	63.6	61.1	53.4	41.9	29.1	14.9		
	DBStd	19.10	10.19	10.23	8.82	5.84	5.88	4.54	3.21	3.13	5.30	6.23	8.24	10.94		
	HDD50	5142	1181	931	619	313	96	5	0	0	27	256	627	1087		
	HDD65	9464	1646	1351	1084	762	513	225	64	126	349	715	1077	1552		
	CDD50	1180	0	0	0	1	48	232	422	343	128	6	0	0		
	CDD65	26	0	0	0	0	0	2	20	4	0	0	0	0		
	CDH74	1895	0	0	0	0	35	485	860	420	94	1	0	0		
	CDH80	299	0	0	0	0	3	67	177	45	7	0	0	0		
	WSAvg	4.9	2.8	3.6	5.2	7.1	6.9	6.2	5.4	4.9	4.7	4.9	3.8	2.9		
	PrecAvg	10.2	0.8	0.7	0.6	0.6	0.7	0.6	1.3	1.5	1.1	0.8	0.6	0.8		
(15) Precipitation	PrecMax	16.5	2.4	2.4	3.3	2.3	1.7	2.5	3.8	3.6	3.3	4.1	1.9	3.1		
	PrecMin	6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.0	0.0	0.0	0.0		
	PrecStd	2.3	0.6	0.5	0.6	0.5	0.5	0.5	0.8	0.8	0.7	0.7	0.4	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	45.4	62.9	67.8	79.0	84.5	87.6	83.2	81.1	71.9	60.1	46.0		
		MCWB	32.3	34.0	40.8	43.8	49.7	53.3	54.6	54.3	53.2	47.0	40.8	34.1		
	2%	DB	36.4	40.2	55.2	63.8	73.1	81.8	83.9	81.3	76.6	67.1	54.8	40.0		
		MCWB	30.2	31.8	38.0	41.6	46.9	51.8	54.1	53.6	51.2	45.2	38.3	31.3		
	5%	DB	32.5	36.6	51.5	61.1	70.4	79.5	81.8	79.1	73.3	63.6	50.4	36.2		
		MCWB	28.3	30.5	36.7	40.6	45.9	50.6	53.8	53.2	49.5	43.4	36.8	29.6		
	10%	DB	28.3	33.6	45.8	56.9	66.6	76.8	79.5	75.4	71.7	59.9	45.2	32.0		
		MCWB	24.8	28.7	34.4	38.8	44.7	49.6	54.0	53.2	49.3	42.1	33.8	27.5		
	0.4%	WB	34.2	35.3	41.7	45.0	50.9	56.1	59.7	59.2	56.0	50.0	42.1	35.6		
		MCDB	39.4	41.0	58.9	63.7	74.4	77.6	76.6	74.7	72.8	63.2	56.5	43.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.1	33.0	39.1	42.9	48.5	54.1	58.2	57.4	54.1	47.2	39.3	32.7		
		MCDB	35.4	39.0	54.1	60.4	69.1	74.9	74.4	71.2	69.3	63.8	52.3	38.7		
	5%	WB	28.2	30.9	37.0	41.4	47.1	52.6	57.1	56.2	52.5	45.0	37.1	30.2		
		MCDB	32.1	36.5	48.8	58.3	66.3	73.3	72.6	70.5	67.3	60.6	49.4	35.5		
	10%	WB	25.1	28.4	34.9	39.6	45.6	51.2	56.1	55.1	50.9	43.0	34.4	27.3		
		MCDB	28.8	33.0	45.3	55.6	64.8	72.4	71.4	69.6	65.7	57.7	44.6	31.9		
	5% DB	MDBR	25.5	25.0	25.7	27.7	30.8	34.9	31.9	30.9	33.3	32.2	28.5	25.0		
		MCDBR	25.7	25.9	33.2	37.2	38.4	40.0	36.6	36.1	39.0	40.4	37.2	26.9		
		MCWBR	19.9	18.8	18.1	18.5	17.0	16.3	13.8	14.1	17.4	21.3	22.3	19.0		
		MCDBR	23.9	24.4	30.2	33.8	34.0	34.6	28.4	28.4	30.4	34.7	34.8	24.6		
	5% WB	MCWBR	18.9	17.9	17.1	17.1	15.3	14.2	10.9	11.5	13.7	18.3	21.2	17.8		
(40) Clear-Sky Solar Irradiance	taub		0.223	0.228	0.256	0.287	0.300	0.298	0.334	0.331	0.286	0.261	0.238	0.225		
	taud		2.521	2.529	2.511	2.475	2.512	2.520	2.468	2.455	2.601	2.634	2.604	2.531		
	Ebn at Noon		315	325	322	315	310	309	297	296	308	309	307	306		
	Edh at Noon		23	27	30	34	33	33	35	34	28	24	22	22		
(44) All-Sky Solar Radiation	RadAvg		846	1114	1570	1889	2099	2396	2091	1865	1681	1285	931	732		
	RadStd		51	70	123	110	161	150	121	86	100	99	64	53		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+22

Nomenclature: See separate page